

Q1 2021 (27 Aug Shift 1)

Let $\frac{\sin A}{\sin B} = \frac{\sin(A-C)}{\sin(C-B)}$, where A, B, C are angles of a triangle ABC. If the lengths of the sides opposite these

angles are a, b, c respectively, then :

(1) $b^2 - a^2 = a^2 + c^2$

(2) b^2, c^2, a^2 are in A.P.

(3) c^2, a^2, b^2 are in A.P.

(4) a^2, b^2, c^2 are in A.P.

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Answer Key

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Q1 (2)

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Q1 (2)

$$\frac{\sin A}{\sin B} = \frac{\sin(A-C)}{\sin(C-B)}$$

As A, B, C are angles of triangle

$$A + B + C = \pi$$

$$A = \pi - (B + C)$$

$$\text{So, } \sin A = \sin(B + C) \dots (1)$$

$$\text{Similarly } \sin B = \sin(A + C) \dots (2)$$

From (1) and (2)

$$\frac{\sin(B+C)}{\sin(A+C)} = \frac{\sin(A-C)}{\sin(C-B)} \quad \sin(C+B) \cdot \sin(C-B) = \sin(A-C) \sin(A+C) \quad \sin^2 C - \sin^2 B = \sin^2 A - \sin^2 C$$

$$\{\because \sin(x+y) \sin(x-y) = \sin^2 x - \sin^2 y\} \quad 2 \sin^2 C = \sin^2 A + \sin^2 B$$

By sine rule

$$2c^2 = a^2 + b^2$$

$$\Rightarrow b^2, c^2 \text{ and } a^2 \text{ are in A.P.}$$